

ELECT-SENTINEL OSINT

Automated Global Election Disinformation & Public Confusion Intelligence Platform

Topic: Topic 1 — Election Disinformation & Confusion OSINT

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GitHub Repository:

<https://github.com/being-cheema/elect-sentinel-osint>

Roll Number: 23011103011

Operational Status: 100% Real Live Global Digital Streams

Date: August 20, 2026

1. Executive Summary & Problem Statement

During election periods, massive volumes of publicly accessible digital content are disseminated across international news websites, discussion forums, social media channels, and messaging boards. Some content contains targeted misleading claims, voter suppression tactics, fabricated rumors regarding polling procedures or equipment, and coordinated messaging designed to confuse voters and delegitimize electoral processes. Manually monitoring these heterogeneous sources is practically impossible due to the volume, velocity, and global dispersion of digital communication.

The Solution: **ELECT-SENTINEL OSINT** is an enterprise-grade automated threat intelligence platform engineered to continuously ingest, analyze, cluster, verify, and visualize election-related digital content in real time from across the globe. The system operates on **100% real live data**, resolving geographic origins across 60+ countries and providing election security analysts with an 8-workstation Cyber Operations Center to identify emerging storylines, track viral lifecycles, and synthesize rapid counter-disinformation briefings.

2. System Architecture & Core Capabilities

Component Layer	Technical Implementation & Responsibilities
Multi-Source Live Ingestion	Multi-threaded ingestion engine (12 worker threads) connecting to 25+ global feeds: international fact-checkers (EUvsDisinfo, PolitiFact, FactCheck.org, FullFact UK), worldwide broadcasters (BBC, Euronews, DW, Al Jazeera, France24, The Guardian), regional wires (The Hindu, Times of India, Daily Maverick, MercoPress), and decentralized Mastodon social firehoses.
NLP & Confusion Classifier	Multi-category threat engine classifying: (1) <i>Voter Suppression</i> , (2) <i>Integrity & Tampering</i> , (3) <i>Synthetic Deepfakes</i> , (4) <i>Official Impersonation</i> , (5) <i>Premature Results</i> , (6) <i>Voter Intimidation</i> , (7) <i>Legitimate News</i> . Computes composite Confusion Threat Score (0–100), Epistemic Uncertainty Index, Urgency/Outrage markers, and Bot Inauthenticity Probability.
Dynamic Topic Clustering	TF-IDF vectorization and cosine similarity clustering to automatically aggregate discrete social posts and news articles into active Emerging Narratives , tracking lifecycles: <i>Emerging</i> → <i>Accelerating</i> → <i>Critical Peak</i> → <i>Debunked</i> → <i>Dormant</i> .
Statutory Ground Truth Engine	Automated cross-verification of online claims against verified statutory election regulations (polling hours, air-gapped machine mandates, paper audit trails, ID laws) to immediately detect factual contradictions and auto-synthesize rapid rebuttal text.
Global Geographic Resolver	Entity extraction dictionary resolving content from 60+ countries across the Americas, Europe, Asia-Pacific, Africa, and Middle East into exact latitude/longitude coordinates for live geospatial heatmapping.
Network & CIB Engine	NetworkX graph modeling entities, narratives, accounts, and hashtags to detect Coordinated Inauthentic Behavior (CIB) swarms, amplification hubs, and cross-platform spread topology.

3. Directory of 25+ Connected Global Live Sources

The platform ingests from genuine, open-source live digital feeds around the clock with an autonomous background worker polling every 25 seconds:

Source Category	Key Connected Outlets & Endpoints	Geographic Scope
Disinfo & Fact Checks	EUvsDisinfo Database, PolitiFact Truth-O-Meter, FactCheck.org, FullFact UK, Google News Global Disinformation & Deepfakes Monitor	Global / International
Global Broadcasters	BBC World News, Euronews International, The Guardian World, Al Jazeera English, Deutsche Welle (DW), France24 International	Worldwide / Multi-lingual
Asia-Pacific Wires	The Hindu, Times of India, NDTV India, Google News Australia (Australian Electoral Commission & Federal Politics)	India, South Asia, Australia
Americas & Africa	Google News Canada, Google News US, Daily Maverick (South Africa), AllAfrica (40+ nations), MercoPress (Latin America / Mercosur)	Americas & Pan-Africa

Decentralized Social	Mastodon Public Federation Live Firehose (15+ international tags: #election, #voting, #democracy, #wahl, #elecciones, #eleicoes, #disinformation)	Global Federated Nodes
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4. Cyber Operations Center & Analyst Workspaces

The user interface is an elite dark-mode Cyber Operations Center featuring 8 specialized analyst workstations:

- 1. Threat Radar:** Live DEFCON threat indicator (Critical / Elevated / Nominal), real-time KPI telemetry dials, Category Threat Distribution Chart (Chart.js doughnut), Dissemination Bar Chart, and incoming real-time global stream ticker with audio chimes.
- 2. Narrative Intelligence:** Dynamically clusters posts into evolving storylines. Displays narrative lifecycle badges (Emerging, Accelerating, Critical Peak, Debunked), viral velocity (posts/hr), cross-platform footprint, and drill-down links.
- 3. Analyst Triage Queue:** Filterable live investigation queue with multi-dimensional selectors (Category, Platform, Priority P0-P3, Status, Min Confusion slider) and search bar. Features deep forensic inspector drawer modal with SHA-256 evidence fingerprints and status updates.
- 4. Actor & Propagation Graph:** Interactive physics-based HTML5 Canvas network simulation with spring-repulsion dynamics, zoom/pan controls, draggable nodes, and community cluster detection for Coordinated Inauthentic Behavior (CIB) bot rings.
- 5. Global Geospatial Map:** Interactive map powered by Leaflet.js (CartoDB Dark Matter styling). Renders localized threat circles dynamically sized and colored by confusion index across 60+ countries worldwide.
- 6. Live OSINT Scanner:** Ad-hoc forensic investigation tool allowing analysts to paste any live URL, tweet/post, or unverified claim from any country to run instant linguistic pattern extraction, uncertainty scoring, and country resolution.
- 7. Ground Truth Knowledge Base:** Curated repository of official statutory rules and voting security mandates with one-click copyable fact-check rebuttal templates for rapid counter-messaging.
- 8. Intelligence Dossiers & Reports:** Formal Case Management system enabling analysts to bundle evidence and synthesize publication-ready OSINT Threat Intelligence Briefings exportable to Markdown, HTML, and printable formats.

5. Technical Implementation & Technology Stack

Backend Stack	Python 3.9+, FastAPI, Uvicorn, SQLite3 (zero external database dependency), NetworkX (graph topology & community detection), Scikit-Learn (TF-IDF & Cosine Clustering), Feedparser, Requests, ThreadPoolExecutor (12 concurrent worker threads), Pydantic, Server-Sent Events (SSE).
Frontend Stack	Modern Vanilla JavaScript (SPA architecture), Custom Cyber Defense Dark-Mode CSS Design System (Glassmorphism, responsive grid), HTML5 Canvas Physics Engine, Leaflet.js (Geospatial maps), Chart.js (Telemetry analytics), Google Fonts (Inter & JetBrains Mono), Web Audio API.
Automated Testing	Full Python unit test suite (tests/test_osint_engine.py) testing NLP classification, ground truth contradiction detection, narrative clustering, network graph generation, and global location resolution. 100% Pass Rate (8/8 tests passed in 0.22s).

6. Installation & Execution Guide for Evaluators

To inspect and run the project locally from the GitHub repository:

```
# 1. Clone the GitHub repository
git clone https://github.com/being-cheema/elect-sentinel-osint.git
cd elect-sentinel-osint

# 2. Run the one-click launch script (creates venv, installs dependencies, starts server)
chmod +x run.sh
./run.sh

# 3. Access the Live Cyber Operations Center in your browser
http://localhost:8000

# 4. Run automated test suite
./venv/bin/python -m unittest tests/test_osint_engine.py
```

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Submitted for Academic & Technical Evaluation. System is fully operational and continuously monitoring global live digital streams.